

Home Assistant Help Document

Getting Started

Home Assistant is running as a Docker container on the Mac Mini and can be accessed through a browser on the local network or remotely through Tailscale. The configuration files are stored locally on the Mac Mini in the "homeassistant" directory, which is where automations, dashboard layouts, and integration settings are all kept. Getting familiar with that directory early on is a good idea since most changes to the system at a deeper level will involve files in there.

Tailscale and Remote Access

The incoming team will need to set up their own Tailscale access to connect to the system remotely. A step-by-step guide for getting that set up is included in Hardware / VPN help document we will also be providing. Once connected, the Home Assistant dashboard is accessible from any device on the tailnet when connected to Tailscale.

Adding and Managing Integrations

Integrations are managed through the Settings menu in Home Assistant under the Integrations section. From there, new integrations can be added using the "Add Integration" button, which pulls from a large built-in library of supported devices and services. All of the integrations currently in use in EcoNest were available natively within Home Assistant, so no custom code was needed to connect them.

One thing worth knowing upfront is that some integrations are still in testing or limited-access phases on their respective platforms. Google Nest and Aladdin Connect, in particular, required the account holder to register as a developer before the integration could be activated. This process can be a bit tedious depending on how quickly the platform approves access, just want you to be aware of that.

Additionally, we found with one integration, namely, TP-Link for the Kasa Smart Plugs that we ended up not using had a hardware update and was no longer compatible with Home Assistant. We did not find that to be true with any of the other devices or integrations that we used, but that is another to be aware of when they are hardware/software updates with the devices.

Building and Modifying Dashboards

The EcoNest dashboards were built using HACS, the Home Assistant Community Store, which adds community-built components on top of what Home Assistant provides out of the box. The components currently installed are Mushroom, Button Card, Card Mod, Auto Entities, and Stack in Card. These can be managed through the HACS section in the Home Assistant sidebar.

Dashboard layouts are configured in YAML, which works similarly to HTML and CSS in terms of how elements are nested and styled. Edits can be made either through the raw configuration editor for direct YAML access or through the visual editor built into Home Assistant for simpler changes. Both work fine, and the team used a mix of both throughout the project.

Here are some resources that I found helpful during setup and using HACS as well.

- HACS Getting Started: <https://hacs.xyz>
- HACS Basic Configuration: <https://hacs.xyz/docs/use/configuration/basic/>

There is a large community around Home Assistant and HACS specifically, and most questions about specific components or dashboard layouts have already been answered somewhere in the community forums or documentation.

Below is a brief description of each HACS component currently installed and what it is used for in the EcoNest dashboard. Each of these can be found and installed by searching the component name directly in the HACS store (there is a HACS sidebar) within Home Assistant.

Mushroom: the primary card layout component used throughout the dashboard. It provides clean, modern cards for displaying entity states and is the foundation for most of what you see across all four dashboard views. It is highly customizable and has good documentation on its GitHub page.

Auto Entities: automatically populates a card with entities based on filters you define, such as all devices in a specific room or all entities of a certain type. This is what keeps the room-based cards in the Overview dashboard up to date without having to manually add each device every time something new is integrated.

Card Mod: allows custom CSS styling to be applied directly to individual cards. This is what was used to fine tune the visual appearance of cards beyond what the default options allow, things like adjusting border radius, spacing, icon sizing, and colors.

Stack in Card: groups multiple cards together into a single contained unit. This is useful for keeping related controls visually together on the dashboard without them appearing as separate floating cards.

Slider Button Card: adds interactive slider controls to the dashboard. In EcoNest this is used for brightness sliders on the lighting controls and sensitivity sliders for the sound sensor threshold settings.

A Note on AI Pipeline Integration

As the AI reasoning pipeline continues to be developed, one option worth keeping in mind is the ability to embed custom web content directly into a Home Assistant dashboard using an iframe card. This would allow a custom web UI, such as a panel showing AI outputs, recommendations, or reasoning logs, to be surfaced as its own dashboard tab alongside the

existing Overview, Control Center, Lights, and Energy views. This is not necessarily the direction the project needs to go, and it may require some additional work depending on how the pipeline's frontend is structured. Still it is a relatively straightforward way to bring AI-generated insights into the same interface the household is already using without having to build a completely separate application. It is something to keep in mind as the pipeline matures and the team starts thinking about how those outputs should be presented to the user. We have them currently set up to be SMS messages, but this could be an additional way for the users to look back at all of the messages sent over the last week, month, etc.